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COMP20360 Formal Foundations 2

Discrete Mathematics Theorem Sheet

LAWS OF THE PREDICATE CALCULUS

0*	$\equiv$ associative	$[(X \equiv (Y \equiv Z)) \equiv ((X \equiv Y) \equiv Z)]$
1*	$\equiv$ symmetric	$[X \equiv Y \equiv Y \equiv X]$
2*	$\equiv$ identity	$[X \equiv \text{true} \equiv X]$
3	$\equiv$ reflexive	$[X \equiv X]$
4	<i>true</i>	$[\text{true}]$
5*	$\vee$ symmetric	$[X \vee Y \equiv Y \vee X]$
6*	$\vee$ associative	$[X \vee (Y \vee Z) \equiv (X \vee Y) \vee Z]$
7*	$\vee$ idempotent	$[X \vee X \equiv X]$
8*	$\vee/\equiv$	$[X \vee (Y \equiv Z) \equiv X \vee Y \equiv X \vee Z]$
8a	$\vee/\equiv \equiv$	$[X \vee (Y \equiv Z \equiv W) \equiv X \vee Y \equiv X \vee Z \equiv X \vee W]$
9	$\vee/\vee$	$[X \vee (Y \vee Z) \equiv (X \vee Y) \vee (X \vee Z)]$
10	$\vee$ zero	$[X \vee \text{true} \equiv \text{true}]$
11*	Golden Rule	$[X \wedge Y \equiv X \equiv Y \equiv X \vee Y]$
12	$\wedge$ symmetric	$[X \wedge Y \equiv Y \wedge X]$
13	$\wedge$ associative	$[X \wedge (Y \wedge Z) \equiv (X \wedge Y) \wedge Z]$
14	$\wedge$ idempotent	$[X \wedge X \equiv X]$
15	$\wedge$ identity	$[X \wedge \text{true} \equiv X]$
16	absorption0	$[X \wedge (X \vee Y) \equiv X]$
17	absorption1	$[X \vee (X \wedge Y) \equiv X]$
18	$\vee/\wedge$	$[X \vee (Y \wedge Z) \equiv (X \vee Y) \wedge (X \vee Z)]$
19	$\wedge/\vee$	$[X \wedge (Y \vee Z) \equiv (X \wedge Y) \vee (X \wedge Z)]$
20	$\wedge$ over $\equiv$	$[X \wedge (Y \equiv Z) \equiv X \wedge Y \equiv X \wedge Z \equiv X]$
21	$\wedge/\equiv \equiv$	$[X \wedge (Y \equiv Z \equiv W) \equiv X \wedge Y \equiv X \wedge Z \equiv X \wedge W]$
22	strong MP	$[X \wedge (X \equiv Y) \equiv X \wedge Y]$
23	replacement	$[(X \equiv Y) \wedge (W \equiv X) \equiv (X \equiv Y) \wedge (W \equiv Y)]$
24*	$\Rightarrow$ definition	$[X \Rightarrow Y \equiv X \vee Y \equiv Y]$
25	$\Rightarrow$ reflexive	$[X \Rightarrow X]$
26	$\Rightarrow$ true	$[X \Rightarrow \text{true}]$
27	$\Rightarrow \vee$	$[X \Rightarrow X \vee Y]$
28	$\wedge \Rightarrow$	$[X \wedge Y \Rightarrow X]$
29	shunting	$[X \wedge Y \Rightarrow Z \equiv X \Rightarrow (Y \Rightarrow Z)]$
30	$\Rightarrow$ to $\wedge \equiv$	$[X \Rightarrow Y \equiv X \wedge Y \equiv X]$
31	$\Rightarrow$ over $\equiv$	$[X \Rightarrow (Y \equiv Z) \equiv X \wedge Y \equiv X \wedge Z]$
32*	$\Leftarrow$ definition	$[X \Leftarrow Y \equiv X \wedge Y \equiv Y]$
33	$\Leftarrow \Rightarrow$	$[X \Leftarrow Y \equiv Y \Rightarrow X]$
34*	<i>false</i> definition	$\text{false} \equiv \neg \text{true}$

LAWS OF THE PREDICATE CALCULUS

35*	$\neg$ over $\equiv$	$[\neg(X \equiv Y) \equiv \neg X \equiv Y]$
36	neg-identity	$[X \equiv \text{false} \equiv \neg X]$
37	$\neg$ flip	$[\neg X \equiv Y \equiv X \equiv \neg Y]$
38	$\neg \neg$	$[\neg \neg X \equiv X]$
39*	excluded middle	$[X \vee \neg X]$
40	$\neg \vee \equiv$	$[\neg X \vee Y \equiv X \vee Y \equiv Y]$
41	$\vee$ identity	$[X \vee \text{false} \equiv X]$
42	$\wedge$ zero	$[X \wedge \text{false} \equiv \text{false}]$
43	deMorgan0	$[\neg X \vee \neg Y \equiv \neg(X \wedge Y)]$
44	$\neg \wedge \equiv$	$[\neg X \wedge Y \equiv X \wedge Y \equiv \neg Y]$
45	deMorgan1	$[\neg X \wedge \neg Y \equiv \neg(X \vee Y)]$
46	complement0	$[X \vee (\neg X \wedge Y) \equiv X \vee Y]$
47	complement1	$[X \wedge (\neg X \vee Y) \equiv X \wedge Y]$
48	$\equiv$ to $\wedge \vee$	$[X \equiv Y \equiv (X \wedge Y) \vee (\neg X \wedge \neg Y)]$
49	contradiction	$[X \wedge \neg X \equiv \text{false}]$
50	$\Rightarrow$ to $\neg \vee$	$[X \Rightarrow Y \equiv \neg X \vee Y]$
51	contrapositive	$[X \Rightarrow Y \equiv \neg Y \Rightarrow \neg X]$
52	false $\Rightarrow$	$[\text{false} \Rightarrow X]$
53	reductio ad absurdum	$[X \Rightarrow \text{false} \equiv \neg X]$
54	$\neg$ false	$\neg \text{false} \equiv \text{true}$
55	MP	$[X \wedge (X \Rightarrow Y) \Rightarrow Y]$
56	mutual $\Rightarrow$	$[(X \Rightarrow Y) \wedge (Y \Rightarrow X) \equiv X \equiv Y]$
57	$\vee$ max	$[(X \Rightarrow Z) \wedge (Y \Rightarrow Z) \equiv X \vee Y \Rightarrow Z]$
58	$\wedge$ min	$[(X \Rightarrow Y) \wedge (X \Rightarrow Z) \equiv X \Rightarrow Y \wedge Z]$
59*	$\neq$ definition	$[(X \neq Y) \equiv \neg(X \equiv Y)]$
60	$\neq$ associative	$[((X \neq Y) \neq Z) \equiv (X \neq (Y \neq Z))]$
61	$\neq$ and $\equiv$ associate	$[((X \neq Y) \equiv Z) \equiv (X \neq (Y \equiv Z))]$
62	$\neq$ and $\equiv$ swap	$[X \equiv Y \neq Z \equiv X \neq Y \equiv Z]$
63	$\neq$ symmetric	$[X \neq Y \equiv Y \neq X]$
64	$\neq$ identity	$[X \neq \text{false} \equiv X]$
65	$\wedge / \neq$	$[X \wedge (Y \neq Z) \equiv X \wedge Y \neq X \wedge Z]$
66	$\Rightarrow$ transitive	$[(X \Rightarrow Y) \wedge (Y \Rightarrow Z) \Rightarrow (X \Rightarrow Z)]$
67	$\vee$ monotonic	$[(X \Rightarrow Y) \Rightarrow (X \vee Z \Rightarrow Y \vee Z)]$
68	$\wedge$ monotonic	$[(X \Rightarrow Y) \Rightarrow (X \wedge Z \Rightarrow Y \wedge Z)]$
69	$\Rightarrow$ antimonotonic in antecedent	$[(X \Rightarrow Y) \Rightarrow ((Y \Rightarrow Z) \Rightarrow (X \Rightarrow Z))]$
70	$\equiv$ transitive	$[(X \equiv Y) \wedge (Y \equiv Z) \Rightarrow (X \equiv Z)]$

LAWS OF THE PREDICATE CALCULUS

100*	<i>trading</i>	$[\langle \forall i: r.i: f.i \rangle \equiv \langle \forall i: \neg r.i \vee f.i \rangle]$
100a	<i>trading</i>	$[\langle \forall i: r.i: f.i \rangle \equiv \langle \forall i: r.i \Rightarrow f.i \rangle]$
101*	$\vee/\vee$	$[X \vee \langle \forall i: r.i: f.i \rangle \equiv \langle \forall i: r.i: X \vee f.i \rangle]$
102*	$\forall/\wedge$	$[\langle \forall i: r.i: f.i \rangle \wedge \langle \forall i: r.i: g.i \rangle \equiv \langle \forall i: r.i: f.i \wedge g.i \rangle]$
103	<i>range split</i>	$[\langle \forall i: r.i: f.i \rangle \wedge \langle \forall i: s.i: f.i \rangle \equiv \langle \forall i: r.i \vee s.i: f.i \rangle]$
104	<i>false range</i>	$[\langle \forall i: false: f.i \rangle \equiv true]$
105*	<i>1-point</i>	$[\langle \forall i: [i=j]: f.i \rangle \equiv f.j]$
106	<i>instantiation</i>	$[\langle \forall i: f.i \rangle \Rightarrow f.j]$
107*	$\exists$ <i>definition</i>	$[\langle \exists i: r.i: f.i \rangle \equiv \neg \langle \forall i: r.i: \neg f.i \rangle]$
108	<i>trading</i>	$[\langle \exists i: r.i: f.i \rangle \equiv \langle \exists i: r.i \wedge f.i \rangle]$
109	$\wedge/\exists$	$[X \wedge \langle \exists i: r.i: f.i \rangle \equiv \langle \exists i: r.i: X \wedge f.i \rangle]$
110	$\exists/\vee$	$[\langle \exists i: r.i: f.i \rangle \vee \langle \exists i: r.i: g.i \rangle \equiv \langle \exists i: r.i: f.i \vee g.i \rangle]$
111	<i>range split</i>	$[\langle \exists i: r.i: f.i \rangle \vee \langle \exists i: s.i: f.i \rangle \equiv \langle \exists i: r.i \vee s.i: f.i \rangle]$
122	<i>false range</i>	$[\langle \exists i: false: f.i \rangle \equiv false]$
123	<i>1-point</i>	$[\langle \exists i: [i=j]: f.i \rangle \equiv f.j]$
124	<i>witness</i>	$[\langle \exists i: r.i: f.i \rangle \Leftarrow f.j]$

Floor/Ceiling Calculus

Floor

$\lfloor \cdot \rfloor$ is a function from real numbers to integers, which satisfies for any reals x and y , and integer n :

- 0* $n \leq \lfloor x \rfloor \equiv n \leq x$
- 1 $\lfloor x \rfloor \leq x$
- 2 $n \leq \lfloor n \rfloor$
- 3 $\lfloor n \rfloor \leq n$
- 4 $n = \lfloor n \rfloor$
- 5 $\lfloor x \rfloor < n \equiv x < n$
- 6 $x < \lfloor x \rfloor + 1$
- 7 $\lfloor x \rfloor \leq x \wedge x < \lfloor x \rfloor + 1$
- 8 $\lfloor x \rfloor \leq \lfloor y \rfloor \Leftarrow x \leq y$
- 9 $\lfloor x \rfloor = n \equiv n \leq x \wedge x < n + 1$
- 10 $\lfloor x \rfloor = n \equiv x - 1 < n \wedge n \leq x$
- 11 $\lfloor x + n \rfloor = \lfloor x \rfloor + n$

Ceiling

$\lceil \cdot \rceil$ is a function from real numbers to integers, which satisfies for any reals x and y , and integer n :

- 12* $\lceil x \rceil \leq n \equiv x \leq n$
- 13 $x \leq \lceil x \rceil$
- 14 $\lceil n \rceil \leq n$
- 15 $n \leq \lceil n \rceil$
- 16 $n = \lceil n \rceil$
- 17 $n < \lceil x \rceil \equiv n < x$
- 18 $\lceil x \rceil - 1 < x$
- 19 $\lceil x \rceil - 1 < x \wedge x \leq \lceil x \rceil$
- 20 $\lceil x \rceil \leq \lceil y \rceil \Leftarrow x \leq y$
- 21 $\lceil x \rceil = n \equiv n - 1 < x \wedge x \leq n$
- 22 $\lceil x \rceil = n \equiv x \leq n \wedge n < x + 1$
- 23 $\lceil x + n \rceil = \lceil x \rceil + n$